of 1653 described the convenient tabular presentation for Binomial coefficient now called Pascal's triangle. Sir Issac Newton generalized the Binomial theorem and made it valid for any rational exponent.

Now we study the **Binomial theorem for $(x + a)^n$**

Theorem(without proof)

If x and 'a' are real numbers, then for all $n \in N$

$$(x+a)^n = nC_0 x^n a^0 + nC_1 x^{n-1} a^1 + nC_2 x^{n-2} a^2 + \ldots + nC_r x^{n-r} a^r + \ldots + nC_{n-1} x^1 a^{n-1} + nC_n x^0 a^n = \sum_{r=0}^{n} nC_r x^{n-r} a^r$$

NOTE

When $n = 0$, $(x+a)^0 = 1$

When $n = 1$,

$(x+a) = 1C_0 x + 1C_1 a = x + a$

When $n = 2$,

$(x+a)^2 = 2C_0 x^2 + 2C_1 x^a + 2C_2 a^2$

$= x^2 + 2xa + a^2$

When $n = 3$,

$(x+a)^3 = 3C_0 x^3 + 3C_1 x^2 a + 3C_2 xa^2 + 3C_3 a^3$

$= x^3 + 3x^2 a + 3xa^2 + a^3$ and so on.

Given below is the Pascal's Triangle showing the co- efficient of various terms in Binomial expansion of $(x+a)^n$

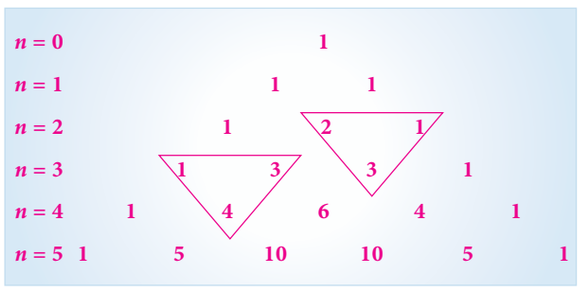


Fig. 2.5

NOTE

(i) Number of terms in the expansion of $(x + a)^n$ is n+1

(ii) Sum of the indices of x and a in each term in the expansion is n

(iii) $nC_0, nC_1, nC_2, nC_3, \ldots nC_r, \ldots nC_n$ are also represented as $C_0, C_1, C_2, C_3, \ldots, C_r, \ldots, C_n$, are called Binomial co-efficients.

(iv) General term in the expansion of $(x+a)^n$ is $t_{r+1} = nC_r x^{n-r} a^r$

(v) Since $nC_r = nC_{n-r}$, for $r = 0, 1, 2, \ldots, n$ in the expansion of $(x + a)^n$, binomial co-efficients of terms equidistant from the beginning and from the end of the expansion are equal.

(vi) Sum of the co-efficients in the expansion of $(1+x)^n$ is equal of 2^n

(vii) In the expansion of $(1+x)^n$, the sum of the co-efficients of odd terms = the sum of the co-efficients of the even terms $= 2^{n-1}$

Middle term of $(x + a)^n$

Case (i) If n is even, then the number of terms $n + 1$ is odd, then there is only one middle term, given by $t_{\frac{n}{2}+1}$

Case (ii) If n is odd, then the number of terms $(n+1)$ is even. Therefore, we have two middle terms given by $t_{\frac{n+1}{2}}$ and $t_{\frac{n+3}{2}}$

Sometimes we need a **particular term** in the expansion of $(x + a)^n$. For this we first write the general term t_{r+1}. The

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